

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q20-1382

Cwec Iola 345 kV  
September 18, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1382. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Cwec Iola 345 kV substation in the Pennsylvania area. The request was received on January 13, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 407 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$88,188,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                     | Contingency                                            | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------------|--------------------------------------------------------|-------------|--------------|----------|
| Vancouver Island-Cwec Iola 138kV                       | Cwec Iola-Firstenergy Ft Martin Power Sta 345kV Line   | 298.0       | 499          | 100.0    |
| Vancouver Island-Wolf Rd 138kV                         | Firstenergy Ft Martin Power Sta-Cwec Iola 345kV        | 289.1       | 510          | 100.0    |
| Wolf Rd-Missouri City 138kV                            | Firstenergy Ft Martin Power Sta-Cwec Iola 345kV line   | 280.6       | 521          | 100.0    |
| Missouri City-Castle Rock 138kV                        | Cwec Iola-Firstenergy Ft Martin Power Sta 345kV        | 273.5       | 531          | 100.0    |
| Castle Rock-Rinn Vly 138kV                             | Cwec Iola-Firstenergy Fort Martin Power Sta 345kV line | 270.0       | 536          | 100.0    |
| Cwec Iola 345-138 kV Autotransformer                   | Cwec Iola-Firstenergy Fort Martin Power Station 345kV  | 263.9       | 563          | 100.0    |
| Rinn Valley-American Riv 138kV                         | Cwec Iola-Firstenergy Fort Martin Power Station 345kV  | 261.7       | 548          | 100.0    |
| Cwec Iola-Firstenergy Fort Martin Power Sta 345kV Line | Cwec Iola-Vancouver Island 138kV L                     | 133.3       | 1,115        | 100.0    |
| Aligator 500-345 kV XFMR                               | Thiele Kaolin Redy Crk-Enka 345kV Line                 | 103.1       | 896          | 7.8      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

#### 4. Required Network Upgrades and Cost Estimates

| Facility                                              | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Vancouver Island-Cwec Iola 138kV                      | Rebuild line with higher-capacity conductor | \$2,914,000   | 100% (\$2,914,000)    |
| Vancouver Island-Wolf Rd 138kV                        | Rebuild line with higher-capacity conductor | \$2,537,000   | 100% (\$2,537,000)    |
| Missouri City-Wolf Rd 138kV                           | Rebuild line with higher-capacity conductor | \$8,359,000   | 100% (\$8,359,000)    |
| Missouri City-Castle Rock 138kV                       | Rebuild line with higher-capacity conductor | \$11,271,000  | 100% (\$11,271,000)   |
| Castle Rock-Rinn Vly 138kV                            | Rebuild line with higher-capacity conductor | \$3,309,000   | 100% (\$3,309,000)    |
| Cwec Iola 345 kV/138 kV Transformer                   | Replace transformer with higher-rated unit  | \$13,619,000  | 100% (\$13,619,000)   |
| Rinn Vly-American River 138kV                         | Rebuild line with higher-capacity conductor | \$3,134,000   | 100% (\$3,134,000)    |
| Cwec Iola-Firstenergy Fort Martin Power Station 345kV | Rebuild line with higher-capacity conductor | \$31,626,000  | 100% (\$31,626,000)   |
| Alligator 500/345kV Auto                              | Replace transformer with higher-rated unit  | \$30,230,000  | 38% (\$11,419,000)    |
|                                                       | Total allocated to project                  |               | \$88,188,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.